Question 1 on Flexible AC Transmission System (FACTS)

a) Let's suppose the transmission routes A and B has reactance of x and $2x$, respectively

$$P_A = \frac{2}{3}P > P_{A-\max}, P_B = \frac{1}{3}P$$

After overloaded route A is upgraded by adding a similar line in parallel

$$X'_A = \frac{x}{2} \Rightarrow P'_A = \frac{4}{5}P, P'_B = \frac{1}{5}P$$

Upgrading the overloaded route could lead to poor utilisation of the other route

[2 marks]

b) Base case without line compensation

Steady-state stability limit

$$P_{max} = \frac{V_1 V_2}{X} = 4.0 \text{ pu}$$

[0.5 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max}}\right) = 0.85 \text{ rad} = 48.59°$$

[0.5 mark]

Critical clearing angle is obtained by equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P \, d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max} \sin\delta - P) \, d\delta$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 1.13 \text{ rad} = \mathbf{65°}$$

[1 mark]

c) With TCSC

i) With the TCSC the steady-state stability limit is

$$P_{max1} = \frac{V_1 V_2}{X(1-k)} = 5.0 \text{ pu}$$

[1 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 0.64 \text{ rad} = 36.87°$$

Model Answers

[1 mark]

Critical clearing angle is

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 1.25 \text{ rad} = \mathbf{71.78}°$
[1 mark]

ii) Under the assumptions in part (b), critical clearing angle depends solely on the initial power angle δ_0 which is a function of $\left(\frac{P}{P_{max1}}\right)$

[1 mark]

If the new power transfer is P'

$$\frac{P'}{P_{max1}} = \frac{P}{P_{max0}}$$

[1 mark]

$$\Rightarrow P' = P \times \frac{P_{max1}}{P_{max0}} = \frac{P}{1-k} = 3.75 \text{ pu}$$

[1 mark]

d) With TCPAR

A TCPAR with ideal phase angle compensation doesn't alter the voltage magnitude and effective line reactance. Hence, the steady-state stability limit would be unaffeced.

[1 mark]

$$P_{max} = \frac{V_1 V_2}{X} = 4.0 \text{ pu}$$

[1 mark]

e) With SVC

Steady-state stability limit without any susceptance limit

$$P_{max2} = \frac{V_1 V_2}{X/2} = 8.0 \text{ pu}$$

Initial power angle $\delta_0$is given by:

$$\delta_0 = 2 \times \sin^{-1}\left(\frac{P}{P_{max2}}\right) = 44.05°$$

[1 mark]

$B_{max} = 5$ p.u.

Model Answers

When the reactive capacity of the SVC is reached

$$\frac{4}{X}\left(1-\cos\frac{\delta_{lim}}{2}\right)=V^2B_{max}\rightarrow\delta_{lim}=2\times\cos^{-1}\left(1-\frac{XB_{max}}{4}\right)=93.14°$$

[1 mark]

Note $\delta_{lim}>90°$ which dictates the following steps

After the reactive capacity of the SVC is reached i.e. for $\delta>\delta_{lim}$ the power angle characteristics is given by

$$P=\frac{4}{X(4-XB_{max})}\sin\delta=P_{max-lim}\sin\delta$$

[1 mark]

$$P_{max-lim}=\frac{4}{0.4(4-0.4\times3.2)}=5.81\text{ p.u.}$$

[1 mark]

$$\delta_{max}=\pi-\sin^{-1}\left(\frac{P}{P_{max-lim}}\right)=148.96°$$

[1 mark]

Equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}}P\,d\delta=\int_{\delta_{cr}}^{\delta_{lim}}\left(P_{max2}\sin\frac{\delta}{2}-P\right)d\delta+\int_{\delta_{lim}}^{\delta_{max}}(P_{max-lim}\sin\delta-P)\,d\delta$$

[1 mark]

$$\delta_{cr}=2\times\cos^{-1}\left[\frac{P(\delta_{max}-\delta_0)+2P_{max2}\cos\frac{\delta_{lim}}{2}-P_{max-lim}(\cos\delta_{lim}-\cos\delta_{max})}{2P_{max2}}\right]=\mathbf{84.68°}$$

[2 marks]

Model Answers

Question 2 on System Operation with Low Inertia

a)

i)

Rate of Change of Frequency (RoCoF), the instant immediately after the system disturbance
System inertia, disturbance
[1 mark]

Frequency Nadir, the instant when the frequency deviation reaches a maximum value
Inertia, PRF, disturbance (EFR, damping)
[1 mark]

Steady-state Frequency, the period after the frequency reaches the steady-state
PFR, disturbance (EFR, damping)
[1 mark]

ii)

$2H\dot{f}(t) = R(t) - \Delta P_L,$
$R(t) = \frac{R}{T_d}t, \ \ if\ t < T_d,$
$\therefore f(t) = \frac{R}{4HT_d}t^2 - \frac{\Delta P_L}{2H}t,$
[1 mark]

$t_n = \frac{\Delta P_L T_d}{R},$
[1 mark]

$f_{nadir} = |\frac{\Delta P_L^2 T_d}{4HR}|,$
[1 mark]

$\therefore HR \geq \frac{\Delta P_L^2 T_d}{4 f_{max}}.$
[1 mark]

iii)

$EFR^2 - 5.5EFR + 2.7 = 0,$

$EFR = 0.545$ GW,
$R = 3.5EFR = 1.9075$ GW
[1 mark]
$g(R, EFR) = (3 - 0.2EFR)R - 3.5(1.8 - EFR)^2,$
$\frac{\partial g}{\partial R} = 3 - 0.2EFR = 2.891,$
$\frac{\partial g}{\partial EFR} = -0.2R + 2 * 3.5(1.8 - EFR) = 8.4035,$
$\frac{\partial R}{\partial EFR} = \frac{\frac{\partial g}{\partial EFR}}{\frac{\partial g}{\partial R}} = 2.907,$

Model Answers

$\frac{\partial R}{\partial EFR} \times \frac{c_{EFR}}{c_R} = \frac{R}{EFR} = 3.5,$
$\frac{c_{EFR}}{c_R} = 1.204.$
[2 marks]

b)

i) BOOKWORK. The swing equation for a synchronous generator relates the changes of its output power to the variation of its frequency of operation. This is why this equation is often used to implement synthetic inertia controllers that calculate how much the power output must be adjusted, *ΔP,* depending on the measured rate of change of the frequency, ie the RoCoF, $\frac{df}{dt}$:

$$\Delta P = -\frac{2H}{f_{nominal}} S_{base} \frac{df}{dt}$$

In this equation, the inertia time, *H*, appears multiplying, which means that the variation of power is directly proportional to this constant.

A basic explanation along the lines presented above will be enough to obtain 3 marks out of 4. For full marks, we'd expect a further justification of the equation explaining the physical meaning of *H* and how it is related to the inertia of a generator.

The swing equation is derived from the equation that relates the rotational acceleration of a mass to the resulting torque applied to it. A change in the torque of a machine causes a deceleration that is proportional to the inertia of the mass:

$$\mathrm{T} = -\mathrm{I_G} \frac{d\omega}{dt}$$

This equation can be written in terms of power by multiplying by the speed in both sides:

$$\mathrm{P} = \mathrm{T}\omega = -\mathrm{I_G}\omega \frac{d\omega}{dt}$$

This is a non-linear differential equation and it is often linearised around the rated frequency of operation assuming that the frequency won't deviate very much from its rated value:

$$\Delta\mathrm{P} = -\mathrm{I_G}\omega_{\mathrm{rated}} \frac{d\omega}{dt}$$

Next, the inertia of the machine can be expressed in terms of kinetic energy stored:

$$\mathrm{E_{rated}} = \frac{1}{2}\mathrm{I_G}\omega_{rated}^2$$

which gives

$$\Delta\mathrm{P} = -\frac{2\,E_{rated}}{\omega_{\mathrm{rated}}} \frac{d\omega}{dt}$$

Further, for a synchronous generator the frequency of its EMF is directly linked to the rotational speed of the generator, and the previous equation can be written as a function of frequency:

$$\Delta\mathrm{P} = -\frac{2\,E}{\mathrm{f_{rated}}} \frac{df}{dt}$$

Also, the kinetic energy at rated speed can be expressed in relative terms of the power rating of the generator, which facilitates the comparison between generators of different ratings:

Model Answers

$$\mathrm{H} \triangleq \frac{E}{\mathrm{S_{rated}}}$$

which gives

$$\Delta \mathrm{P} = -\frac{2\,H}{\mathrm{f_{rated}}} S_{rated} \frac{df}{dt}$$

[4 marks]

ii) This question has two parts:

a) They ask us to calculate the kinetic energy that is stored in the wind turbine. This can be obtained by using the formula:

$$E = \frac{1}{2} I_g \omega^2$$

This gives: E_1=0.5·1.6·10^7·(20·π/30)^2=35.1 MJ and E_2=28.4 MJ.

[2 marks]

b) This question asks us to calculate how much additional power we would be able to deliver to the grid by borrowing kinetic energy moving the speed from ω_1 to ω_2. We can calculate the difference in energy (6.7 MJ) and divide by the time spent in the process (30 s): ΔP=222 kW, which is about 0.074 pu.

[2 marks]

iii) The kinetic energy available in the airflow that interacts with the turbine is proportional to the cube of the wind speed. Typically, wind turbines can be designed to extract around 45% (that's in practice, the theoretical limit given by the simple Betz model is 59.3%). When the wind speed is very high, the power available is very high. However, operating at high wind speeds requires high rotating speeds and high torque, which the turbine has to be designed to withstand. On the other hand, high wind speeds are rare. This means that while a lot of power is available at high wind speeds, there's little energy to be generated over the year due to these high wind speeds and consequently, it isn't worth sizing the turbine to extract optimal power at this speed range. Typically, wind turbines are designed to reduce the power intake and operate at rated torque and rated rotating speed at all speeds between the rated speed and the cut-out speed.

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[2 marks]

Model Answers

Question 3 on Power System State Estimation

a)

i) What are the states in the context of power system state estimation? [2]

Usually node voltage magnitudes (RMS) and AC phase angles. The students are expected to also include OLTC tap positions , phase shifter positions etc. , some times network parameters if the estimation is formulated to seek that.

ii) What drive(s) these states? [2]

Power networks in the steady state have generations matching demands plus losses. Current and power flows in the system because voltage and angle differences between the nodes, so it is the requirement of the current flows that drives the variation in the states.

iii) How does state estimation help to detect stressed system operation? [4]

State estimation is a solved power flow where voltage magnitudes and phase angles are estimated with reasonably acceptable errors. If magnitude of these voltages are not within the set range of operation, system will have voltage stress. When the angle difference is large, the current that flows can be higher than the allowable limit (thermal limit). Once states are known, with the help of power and current calculations, these quantities are calculated. Also, contingency /security analysis are possible when with network topology and parameters, states are known. A series of power flow are conducted to assess the system operating voltage and feeder currents and examine their suitability for secured operation. This is how state estimation helps to detect stressed operation of the system.

iv) How does state estimation differs for unbalanced system from balanced one? [2]

Unbalanced operation results from several reason, unsymmetrical network, unbalanced loads, running a three phase loads from single or two phases etc . All these will lead to different RMS value of voltage and phase angle associated with each phase. This means each phase must be treated concurrently rather than assuming balanced operation across all three phases. The formulation will involve each phase with increase dimensionality. All modern state estimator within the distribution EMS will have three phase formulation.

b)

i) What is the typical measurement sets in power distribution state estimation? [2]

voltage, currents, powers (real and reactive) etc.

ii) How are those measurements classified? [2]

Metered measurement are more accurate measurements and usually within 3% accuracy. These are real measurements. When in a node there is no demand and generations are connected, one can model that as zero current or power injections. These zero injections at the node are treated as 100% accurate measurements, at times known as virtual measurements. Most often than not the individual loads in power distribution are not measured for power and voltage but for energy. These are difficult to know with very good accuracy for power and voltage. Sometimes

Model Answers

some sample measurements from some nodes are replicated. These are highly inaccurate and termed as pseudo measurements.

iii) What is a linear state estimator? [3]

Linear state estimator is one where states and measurements are linearly related . like voltage and angles are directly measured. PMU measurements come with phase angle measurement. The solution for the estimate through WLS is still a gain matrix but sparser and easier to handle. All the nodes may not have angle measurements, so it may not be fully linear. Error processing is required.

iv) What measurement sets can make the linear state estimation possible? [1]

Voltage, angle and tap positions – when part of the measurement sets , one can formulate a linear state estimator.

v) What is the most suitable algorithm in power distribution state estimation and why? [2]

Weighted Least Square (WLS) – as it is consistent and unbiased with relatively best accuracy in the stochastic sense. Even if the error distribution of the measurements do not fit unimodal distribution, through Gaussian mixture model (GMM) extraction , WLS is best applicable .

[20 marks]

Model Answers

Question 4 on Electricity Market and Peer-to-Peer Trading

a)

i) For off-peak period, the demand "D1" is 300 MW; the wind farm "O1" is operating at 60% of its capacity, the power dispatch of the wind farm is $300 \times 60\% = 180$ MW. Then, the supply-demand curve can be constructed as follow:

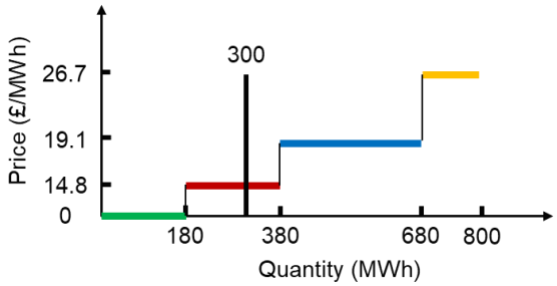


Figure 1: Market demand-supply curve for perfect competition case at off-peak period
[1 mark]

Generation dispatches of 4 units:
Generation of S1 = 120 MW
Generation of S2 = 0 MW
Generation of O1 = 180 MW
Generation of O2 = 0 MW
Market-clearing price = cost of unit S1 = 14.8 £/MWh.
[1 mark]

Generation profits of 4 units:
Profit of S1 = 120 MW × (14.8 – 14.8) £/MWh × 1 hour = £0
Profit of S2 = £0
Profit of O1 = 180 MW × (14.8 – 0) £/MWh × 1 hour = £2,664
Profit of O2 = £0
[1 mark]

For peak period, the demand "D2" is 720 MW; the wind farm "O1" is operating at 80% of its capacity, the power dispatch of the wind farm is $300 \times 80\% = 240$ MW. Then, the supply-demand curve can be constructed as follow:

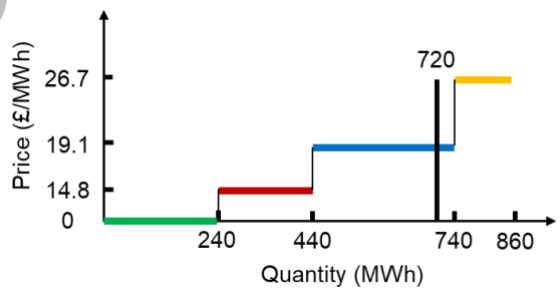


Figure 2: Market demand-supply curve for perfect competition case at peak period
[1 mark]

Generation dispatches of 4 units:
Generation of S1 = 200 MW
Generation of S2 = 280 MW

Model Answers

Generation of O1 = 240 MW
Generation of O2 = 0 MW
Market-clearing price = cost of unit S2 = 19.1 £/MWh.
[1 mark]

Generation profits of 4 units:
Profit of S1 = 200 MW × (19.1 – 14.8) £/MWh × 1 hour = £860
Profit of S2 = 280 MW × (19.1 – 19.1) £/MWh × 1 hour = £0
Profit of O1 = 240 MW × (19.1 – 0) £/MWh × 1 hour = £4,584
Profit of O2 = £0
[1 mark]

ii) Strategic offering: assume that S1 offers strategically at 1.5 times its true cost while S2 continues to offer competitively during the off-peak period; assume that S2 offers strategically at 1.6 times its true cost while S1 continues to offer competitively during the peak period. For both periods, draw market demand-supply curves, calculate generation dispatches, market-clearing prices, and generation profits.

For off-peak hour, S1 offers strategically at 1.5 times its true cost while S2 continues to offer its true cost, so the operational details of the generation units can be updated as:
Table 1: Operational details of generation units

Unit	S1	S2	O1	O2
Technology	Coal	Gas	Wind	Gas
Cost (£/MWh)	**22.2**	19.1	0	26.7
Capacity (MW)	200	300	300	120

Then, the new supply-demand curve can be constructed as follow:

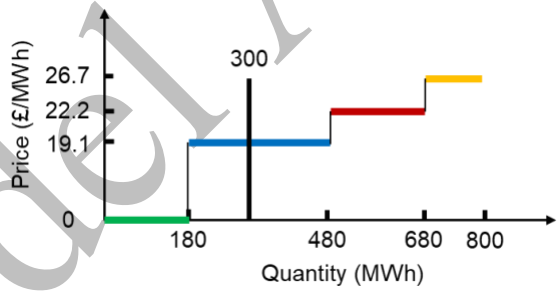


Figure 3: Market demand-supply curve for strategic offering case at off-peak period
[1 mark]

Generation dispatches of 4 units:
Generation of S1 = 0 MW
Generation of S2 = 120 MW
Generation of O1 = 180 MW
Generation of O2 = 0 MW
Market-clearing price = cost of unit S2 = 19.1 £/MWh.
[1 mark]

Generation profits of 4 units:
Profit of S1 = £0
Profit of S2 = 120 MW × (19.1 – 14.8) £/MWh × 1 hour = £516
Profit of O1 = 180 MW × (19.1 – 0) £/MWh × 1 hour = £3,438

Model Answers

Profit of O2 = £0
[1 mark]

For peak hour, S2 offers strategically at 1.6 times its true cost while S1 continues to offer its true cost, so the operational details of the generation units can be updated as:
Table 2: Operational details of generation units

Unit	S1	S2	O1	O2
Technology	Coal	Gas	Wind	Gas
Cost (£/MWh)	14.8	**30.56**	0	26.7
Capacity (MW)	200	300	300	120

Then, the new supply-demand curve can be constructed as follow:

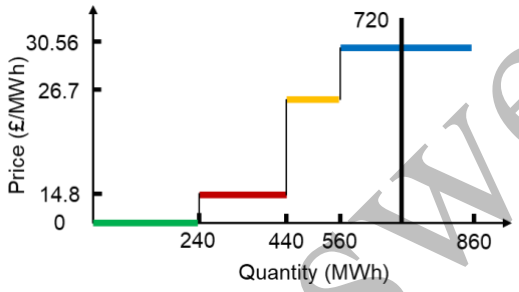


Figure 4: Market demand-supply curve for strategic offering case at peak period
[1 mark]

Generation dispatches of 4 units:
Generation of S1 = 200 MW
Generation of S2 = 160 MW
Generation of O1 = 240 MW
Generation of O2 = 120 MW
Market-clearing price = cost of unit S2 = 30.56 £/MWh.
[1 mark]

Generation profits of 4 units:
Profit of S1 = 200 MW × (30.56 – 14.8) £/MWh × 1 hour = £3,156
Profit of S2 = 160 MW × (30.56 – 19.1) £/MWh × 1 hour = £1,833.6
Profit of O1 = 240 MW × (30.56 – 0) £/MWh × 1 hour = £7,334.4
Profit of O2 = 120 MW × (30.56 – 26.7) £/MWh × 1 hour = £463.2
[1 mark]

b)

i) Based on Table 3, the supply-demand curve of 2 consumers and 2 prosumers for the double-auction market can be constructed as follow:

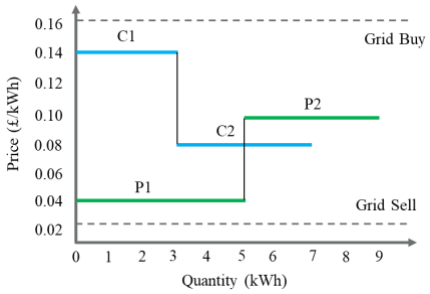


Figure 3: Demand-supply curve for double-auction market.

Model Answers

[1 mark]

First Trading Activity happens between C1 and P1:
Trading price = (0.14 + 0.04)/2 = 0.09 £/kWh
Trading quantity = min(5, 3) = 3 kWh
Updated order book

Consumer/Prosumer	C2	P1	P2
Bid/Offer price (£/kWh)	0.08	0.04	0.10
Quantity (kW)	4	2	4

[1 mark]

Second Trading Activity happens between C2 and P1:
Trading price = (0.08 + 0.04)/2 = 0.06 £/kWh
Trading quantity = min(4, 2) = 2 kWh
Updated order book

Consumer/Prosumer	C2	P2
Bid/Offer price (£/kWh)	0.08	0.10
Quantity (kW)	2	4

[1 mark]

Third Trading Activity happens between C2 and P2:
Trading price = (0.08 + 0.10)/2 = 0.09 £/kWh
Trading quantity = min(2, 4) = 2 kWh
Updated order book

Consumer/Prosumer	P2
Bid/Offer price (£/kWh)	0.10
Quantity (kW)	2

[1 mark]

ii) There are 3 local trading activity in the market, as discussed in (c). The cost of 2 consumers and revenue of 2 prosumers can be calculated as:
Cost of C1 = 3 kWh × 0.09 £/kWh × 1 hour = £0.27
[1 mark]

Cost of C2 = (2 kWh × 0.06 £/kWh + 2 kWh × 0.09 £/kWh) × 1 hour = £0.30
[1 mark]

Revenue of P1 = (3 kWh × 0.09 £/kWh + 2 kWh × 0.06 £/kWh) × 1 hour = £0.39
[1 mark]

Revenue of P2 = 2 kW × 0.09 £/kWh × 1 hour = £0.18
[1 mark]

Model Answers